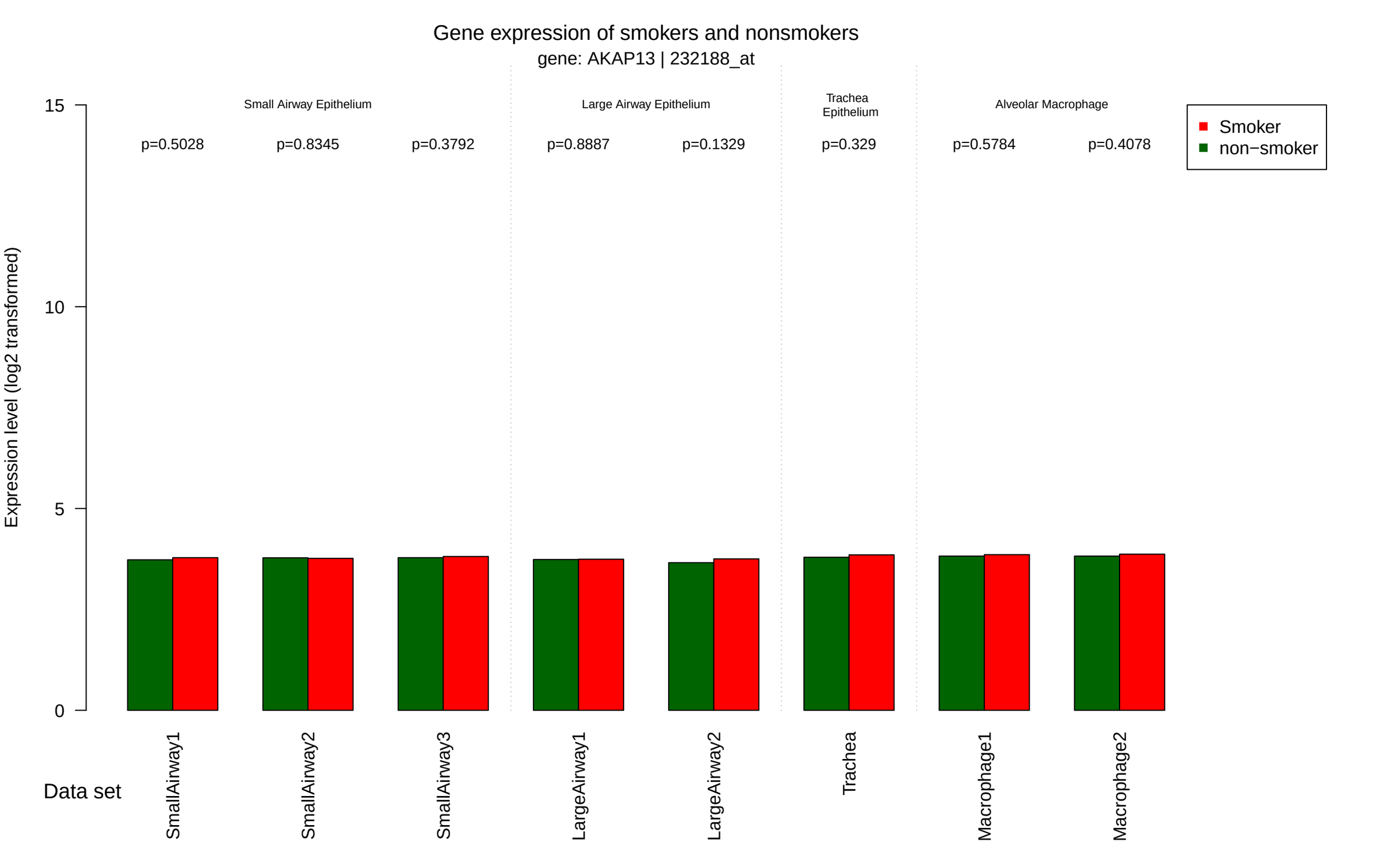
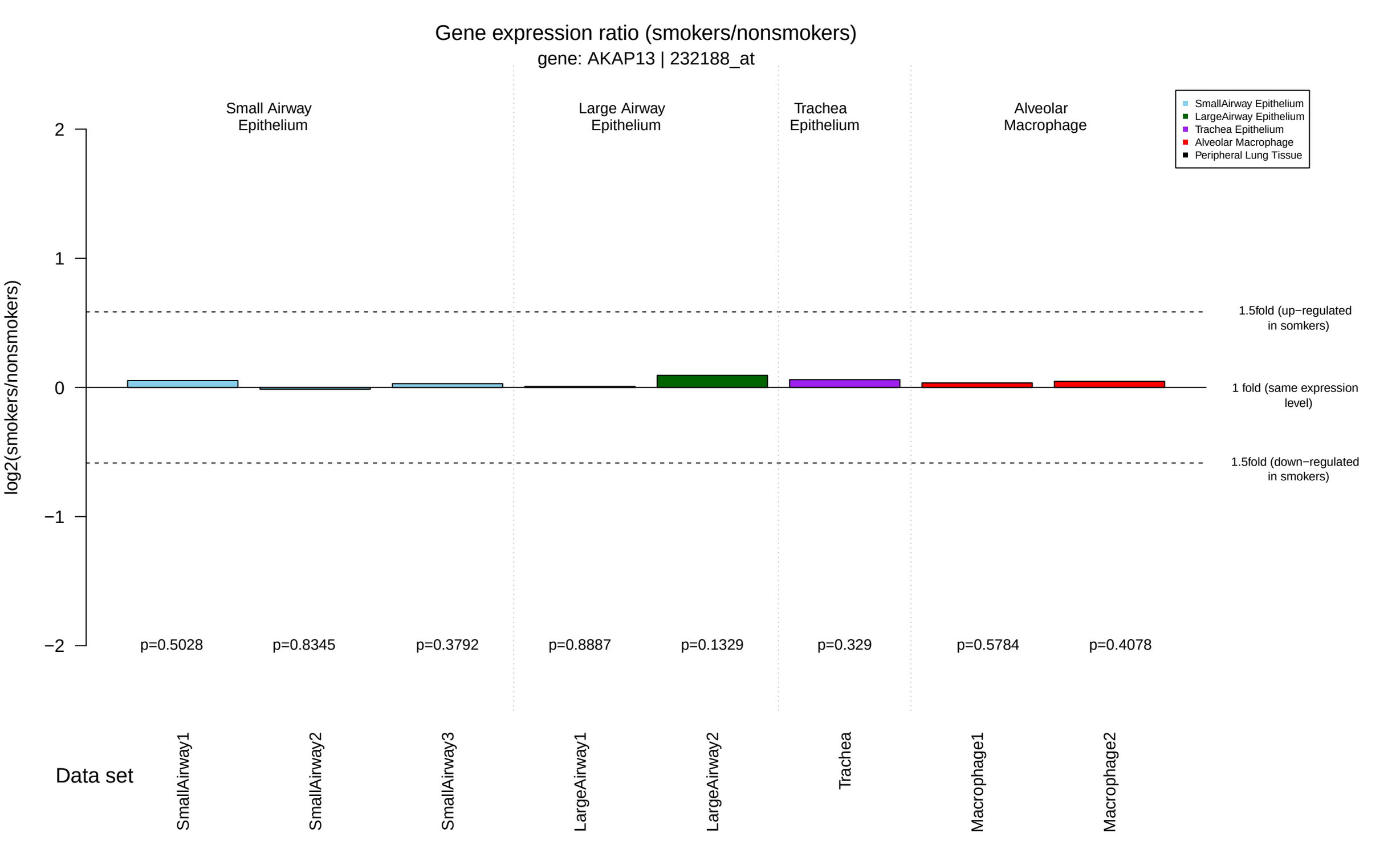
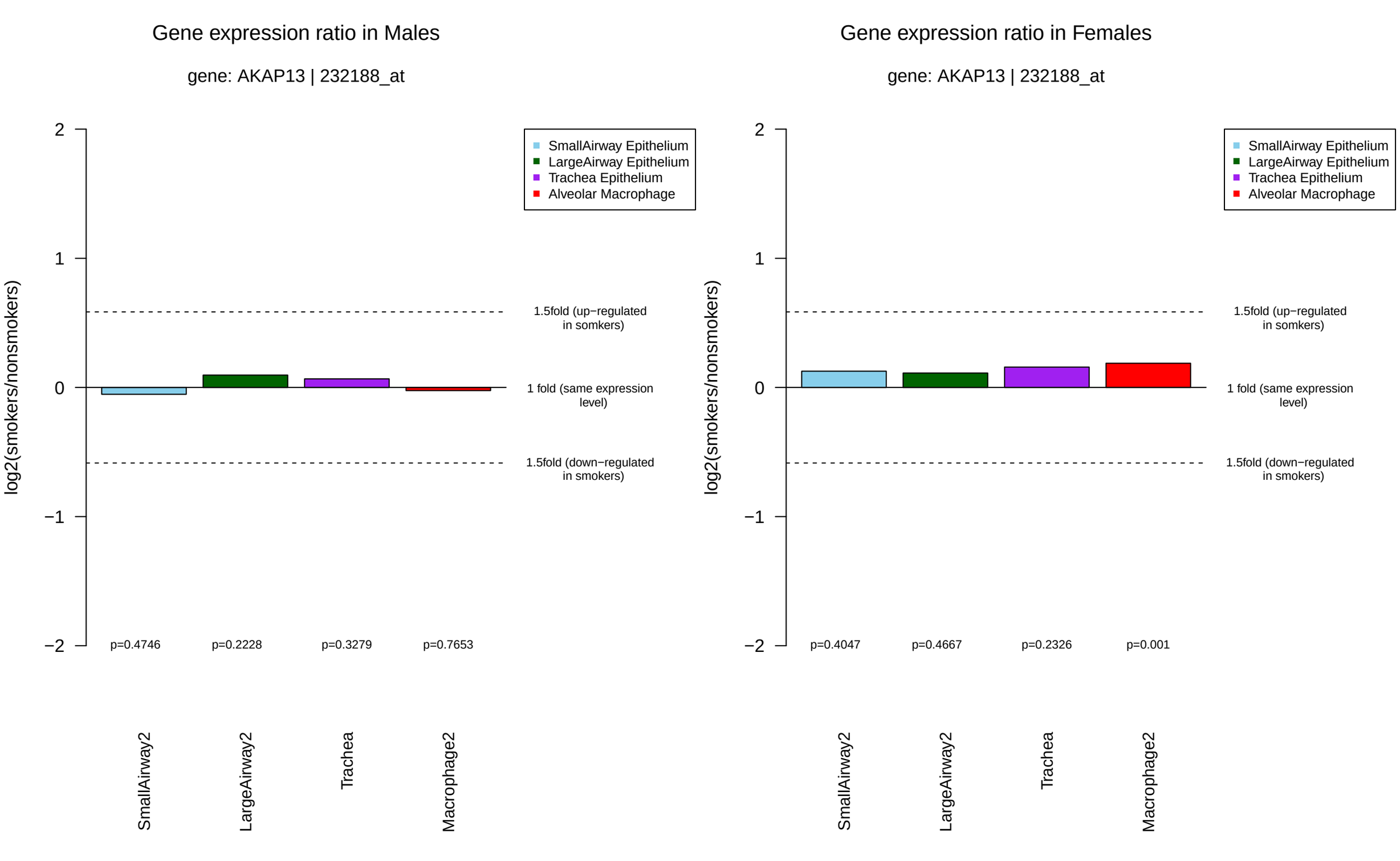
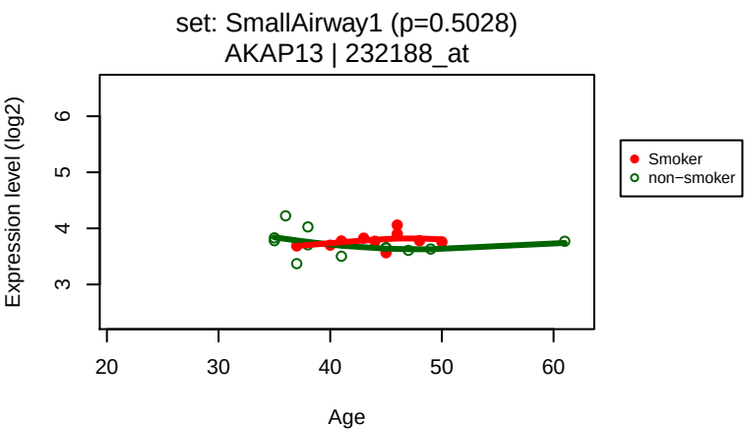
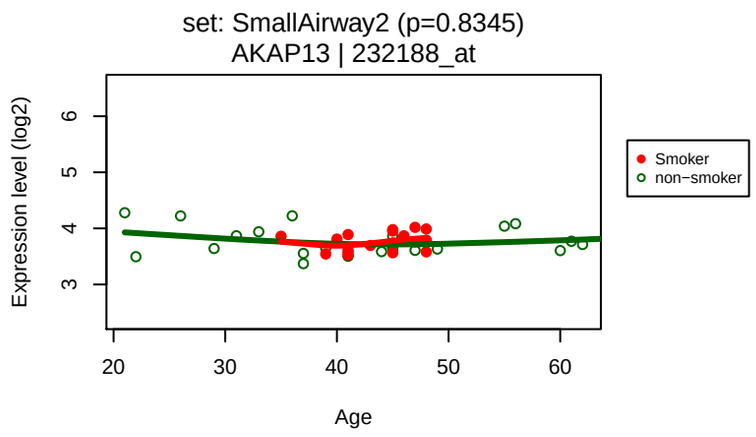
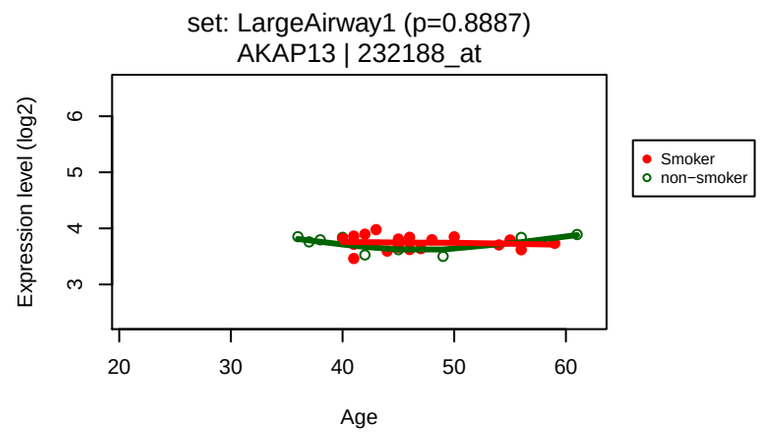

Gene expression of smokers and nonsmokers
gene: AKAP13 | 232188_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5028
p=0.8345
p=0.3792
p=0.8887
p=0.1329
p=0.329
p=0.5784
p=0.4078
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

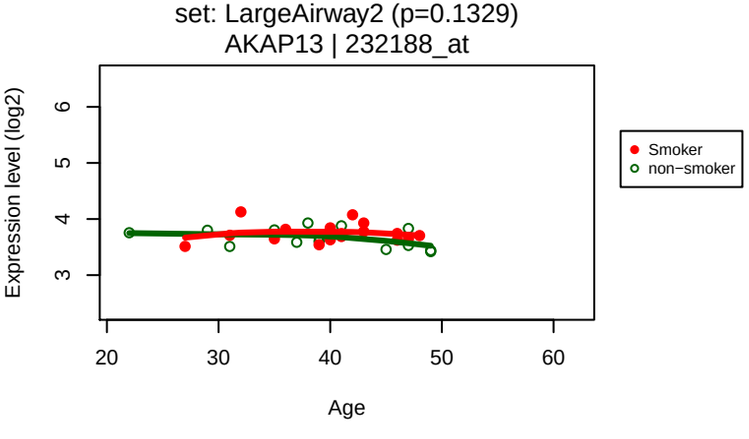

Gene expression ratio (smokers/nonsmokers)
gene: AKAP13 | 232188_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5028
p=0.8345
p=0.3792
p=0.8887
p=0.1329
p=0.329
p=0.5784
p=0.4078
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

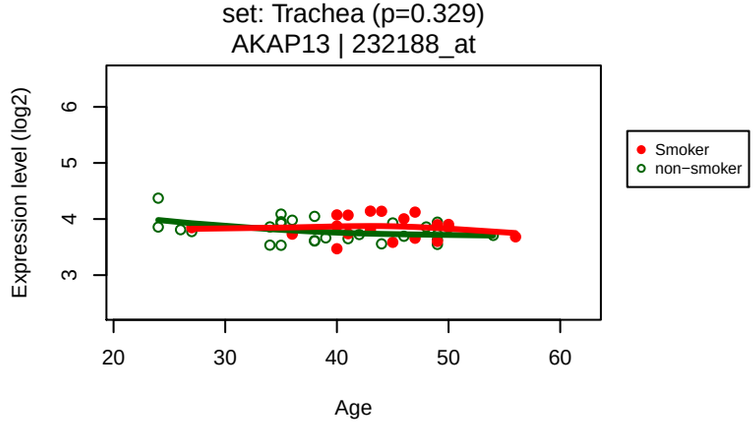

Gene expression ratio in Males
gene: AKAP13 | 232188_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4746
p=0.2228
p=0.3279
p=0.7653
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP13 | 232188_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4047
p=0.4667
p=0.2326
p=0.001
SmallAirway2
LargeAirway2
Trachea
Macrophage2

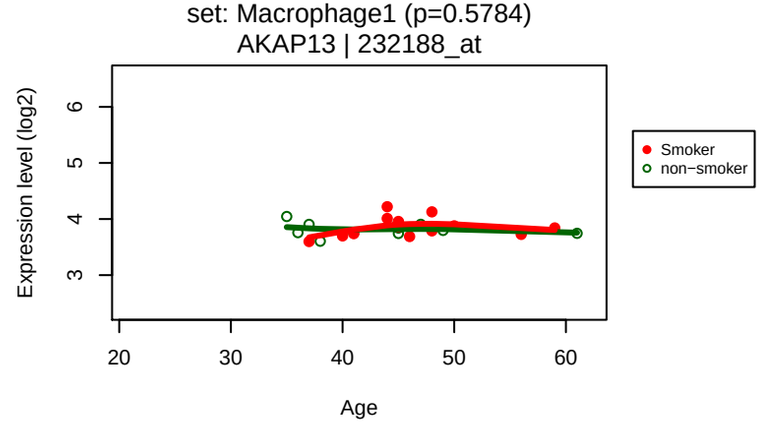

set: SmallAirway1 (p=0.5028)
AKAP13 | 232188_at
Expression level (log2)
Age
Smoker
non−smoker

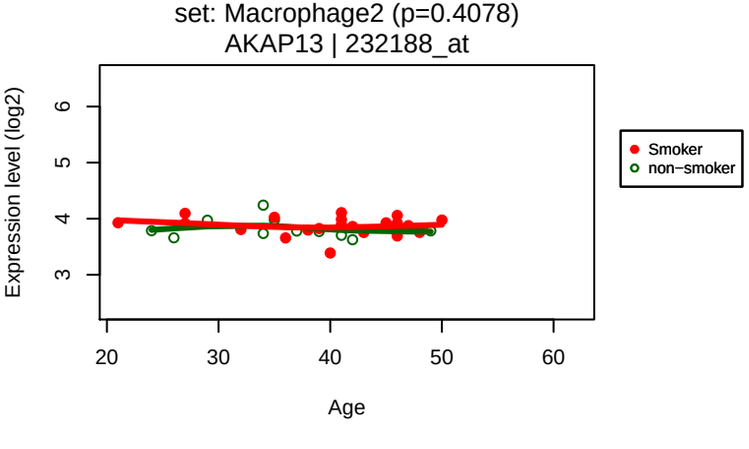

set: SmallAirway2 (p=0.8345)
AKAP13 | 232188_at
Expression level (log2)
Age
Smoker
non−smoker

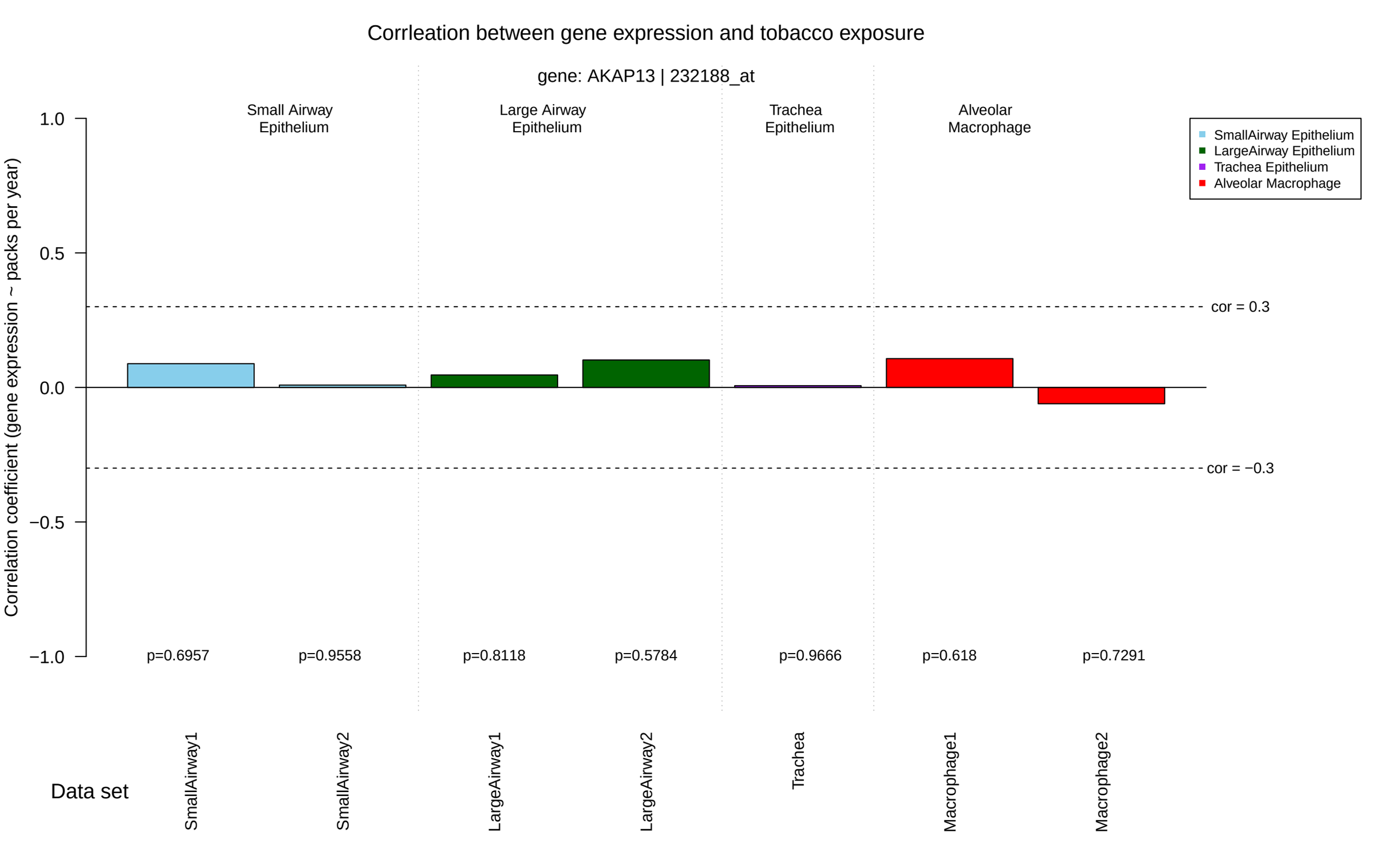

set: LargeAirway1 (p=0.8887)
AKAP13 | 232188_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1329)
AKAP13 | 232188_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.329)
AKAP13 | 232188_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5784)
AKAP13 | 232188_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4078)
AKAP13 | 232188_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP13 | 232188_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6957
p=0.9558
p=0.8118
p=0.5784
p=0.9666
p=0.618
p=0.7291
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2